IMO Training – Number Theory
Order, Primitive Roots and Diophantine Equations

不定方程

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Notes

$3^1 \equiv 3 \ (7)$
$3^2 \equiv 2 \ (7)$
$3^3 \equiv 6 \ (7)$
$3^4 \equiv 4 \ (7)$
$3^5 \equiv 5 \ (7)$
$3^6 \equiv 1 \ (7)$

order of 3 mod 7 is 6
$ord_7(3) = 6$

$3^1 \equiv 3 \ (11)$
$3^2 \equiv 9 \ (11)$
$3^3 \equiv 5 \ (11)$
$3^4 \equiv 4 \ (11)$
$3^5 \equiv 1 \ (11)$

Consider sequence $3^1, 3^2, 3^3, 3^4, \ldots$
this sequence is actually repetition of $3, 9, 5, 4, 1$.

We can only tell $3^{10} \equiv 1 \ (11)$ by F.l.T.
But $3^5 \equiv 1 \ (11)$ already!

Definition: For $(a, m) = 1$, the order of a mod m is the minimum positive integer n such that $a^n \equiv 1 \pmod m$.
$n = ord_m(a)$.

By Euler's Theorem,
$a^{\phi(m)} \equiv 1 \ (m)$.

a is a primitive root of m if $ord_m(a) = \phi(m)$.

1 **Example.** Find $\text{ord}_3(2)$.

2 **Example.** Find $\text{ord}_{17}(3)$.

3 **Example.** Find $\text{ord}_{101}(2)$.

4 **Example.** Find all primitive roots of 29 between 1 and 28. $\phi(29) = 28$ as 29 is prime.

1. $\because 2^1 \equiv 2\ (3)$, $2^2 \equiv 1\ (3)$,
$\therefore \text{ord}_3(2) = 2$.

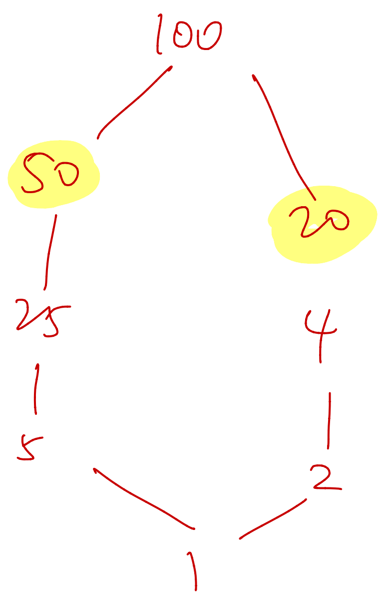
2. $\because 3^1 \equiv 3\ (17)$
$3^2 \equiv 9\ (17)$
$3^4 \equiv 13\ (17)$
$3^8 \equiv -1\ (17)$
$3^{16} \equiv 1\ (17)$

$\therefore \text{ord}_{17}(3) = 16$.

By F.l.T.,
$3^{16} \equiv 1\ (17)$
$\therefore \text{ord}_{17}(3) \mid 16$.

3. By F.l.T., $2^{100} \equiv 1\ (101)$.
So $\text{ord}_{101}(2) \mid 100$.
It suffices to check $2^{50}, 2^{20} \not\equiv 1\ (101)$.

$100 = 2^2 \times 5^2$



Note that $2^{10} \equiv 14\ (101)$.

$2^{20} \equiv (14)^2\ (101)$
$\equiv -6\ (101)$.

$2^{50} \equiv (2^{20})^2 (2^{10})\ (101)$
$\equiv (-6)^2 (14)\ (101) \equiv -1\ (101)$.

$\therefore \text{ord}_{101}(2) = 100$.

Primitive Roots of 29 between 1 and 28.

$28 = 2^2 \times 7$.

<u>Claim</u>: 2 is a primitive root of 29.

<u>Proof</u>: By F.L.T., $2^{28} \equiv 1 (29)$. It suffices to show $2^{14} \not\equiv 1 (29)$ & $2^4 \not\equiv 1 (29)$.

$565 \times 29 - 1$

The latter is obvious. As $2^{14} = 16384 \equiv -1 (29)$.

The claim follows.

$2^1, 2^2, 2^3, \ldots, 2^{28}$ are different numbers mod 29.

order $= 14$, not a primitive root.

$\therefore$ Primitive roots of 29 (mod 29), are those of the form 2^k, where $(k, 28) = 1$, $1 \le k \le 28$.

$2^1, 2^3, 2^5, 2^9,$
$2^{11}, 2^{13}, 2^{15}, 2^{17},$
$2^{19}, 2^{23}, 2^{25}, 2^{27}$

2, 3, 8, 10, 11, 14, 15, 18, 19, 21, 26, 27.

1 **Example.** Prove that if $a, m, n \in \mathbb{N}$, where a, m are coprime, then $a^n \equiv 1 \pmod{m}$ iff $\mathrm{ord}_m(a) \mid n$.

2 **Example.** Prove that for any prime p, every prime divisor of $2^p - 1$ is greater than p.

1. $\Leftarrow$) By definition, write $n = k \cdot \mathrm{ord}_m(a)$.

Then as $a^{\mathrm{ord}_m(a)} \equiv 1 \pmod{m}$,

$$a^n = \left(a^{\mathrm{ord}_m(a)}\right)^k \equiv 1 \pmod{m}. \checkmark$$

$\Longrightarrow$) By Division Algorithm, let $n = q \cdot \mathrm{ord}_m(a) + r$, $q, r \in \mathbb{Z}$, $0 \le r < \mathrm{ord}_m(a)$.

Then for $a^n \equiv 1 \pmod{m}$,

$$a^{q \cdot \mathrm{ord}_m(a) + r} \equiv 1 \pmod{m}.$$

As $a^{\mathrm{ord}_m(a)} \equiv 1 \,(m)$, $a^r \equiv 1 \pmod{m}$.

By minimality of $\mathrm{ord}_m(a)$, $r = 0$.

$\therefore \mathrm{ord}_m(a) \mid n$. $\square$.

2. Consider any prime q s.t. $q \mid 2^p - 1$.

Then $2^p \equiv 1 \ (q)$. So $\text{ord}_q(2) \mid p$.

So $\text{ord}_q(2) = 1$ or p.

If $\text{ord}_q(2) = 1$, then $2^1 \equiv 1 \ (q)$. Cannot be true.

$\therefore \text{ord}_q(2) = p$,

Note that $2^{q-1} \equiv 1 \ (q)$, so $p \mid q-1$.

So $p \leq q-1$, $q > p$. $\square$.

1 **Example.** Let p be an odd prime, and let q, r be primes s.t. $p \mid q^r + 1$. Prove that $2r \mid p-1$ or $p \mid q^2 - 1$.

2 **Example.** Let $a, n \in \mathbb{N}$ with $a > 1$. If p is an odd prime s.t. $p \mid a^{2^n} + 1$, prove that $2^{n+1} \mid p-1$.

1. $\because p \mid q^r + 1$, $\therefore q^r \equiv -1 (p)$.

$q^{2r} \equiv 1 (p)$.

Then $\mathrm{ord}_p(q) \mid 2r$. Since r is prime, $\mathrm{ord}_p(q) = 1, 2, r, 2r$.

if $\mathrm{ord}_p(q) = 1$, then $q \equiv 1 (p)$.

But this cannot be true, because if so $q^r + 1 \equiv 2 (p)$, and $p \neq 2$. #.

if $\mathrm{ord}_p(q) = 2$, then $q^2 \equiv 1 (p)$, and so $p \mid q^2 - 1$.

if $\mathrm{ord}_p(q) = r$, then $q^r \equiv 1 (p)$, and so $2 \equiv 0 (p)$. #

$q^r + 1 \equiv 2 (p)$,

if $\mathrm{ord}_p(q) = 2r$, then by F.L.T., $q^{p-1} \equiv 1 (p)$, so $2r \mid p-1$.

$\therefore p \mid q^2 - 1$ or $2r \mid p-1$.

$\square$

2. $\because p \mid a^{2^n} + 1$, $\therefore a^{2^n} \equiv -1 (p)$.

$(a^{2^n})^2 \equiv 1 (p)$

$a^{2^{n+1}} \equiv 1 (p)$.

$\therefore \mathrm{ord}_p(a) \mid 2^{n+1}$.

But since $a^{2^n} \not\equiv 1 (p)$, $\therefore \mathrm{ord}_p(a) = 2^{n+1}$.

By F.L.T., $a^{p-1} \equiv 1 (p)$. So $2^{n+1} \mid p-1$.

$\square$

Example. (Putnam 1972)Let $n \in \mathbb{Z}, n \geq 2$. Prove that $n \nmid 2^n - 1$.

Example. Prove that for all $a, n \in \mathbb{N}$ with $a > 1$, $n \mid \phi(a^n - 1)$.

1 **Example.** Find all $x, y \in \mathbb{Z}$ s.t. $3x + 4y = 1$.

2 **Example.** Find all $x, y \in \mathbb{Z}$ s.t. $11x + 22y = 13$. mod some no. on both sides.

3 **Example.** Find all $n \in \mathbb{N}$ s.t. $2^8 + 2^{11} + 2^n$ is a perfect square.

x $4\uparrow$ y $\downarrow -3$

1. One of the solutions is $(x, y) = (-1, 1)$.
   In general, $(x, y) = (-1+4k, 1-3k)$, where k is any integer.

2. Consider mod 11, then $0 \equiv 2 \pmod{11}$. Absurd.
   No solution.

3. Let $2^8 + 2^{11} + 2^n = m^2$.
   Then $2304 + 2^n = m^2$.

$$m^2 - 48^2 = 2^n$$
$$(m+48)(m-48) = 2^n.$$

Let $\begin{cases} m+48 = 2^a \\ m-48 = 2^b, \end{cases}$ some +ve integers m, $a > b$, and $a+b=n$.

$$2^a - 2^b = 96.$$

The only solution is $a = 7$, $b = 5$.

$\therefore$ $n = 12$, and $2^8 + 2^{11} + 2^{12} = 6400 = 80^2$

1 **Example.** Find all $x, y \in \mathbb{Z}$ s.t. $x^3 + y^3 = 2006$.

2 **Example.** Find all $x, y \in \mathbb{Z}$ s.t. $y^2 = x^3 + 7$.

1. Consider mod 7. $x^3 \equiv 0, 1, -1\ (7)$

$x^3 + y^3 \equiv -2, -1, 0, 1, 2\ (7)$.

But $2006 \equiv 4 \pmod 7$. So no solution.

Recall Cubing Lemma,

For $p \not\equiv 1(3)$, then every integer is a cube mod p.

For $p \equiv 1(3)$, there are only $\frac{p-1}{3} + 1$ cubes mod p.

$x^3 \equiv 0, 1, -1, 8, -8 \pmod{(3)}$.

2. If x even, then $y^2 \equiv 3 \pmod 4$. $\#$

$\therefore$ x odd, and so y even.

Mod 4, $x^3 \equiv 1(4)$, so $x \equiv 1(4)$.

Then rewrite the equation, $y^2 + 1 = x^3 + 8$.

$$y^2 + 1 = (x+2)(x^2 - 2x + 4).$$

Then $x + 2 \equiv 3(4)$, we can always find a prime

p s.t. $p \mid x+2$ & $p \equiv 3(4)$.

And $p \mid y^2 + 1$ as well.

But such p must be $\equiv 1 \pmod 4$. $\#$

$\therefore$ No solution.

1 **Example.** Find all $x, y \in \mathbb{Z}_{\geq 0}$ s.t. $1 + 3^x = 2^y$. $(0,1)$, $(1,2)$

2 **Example.** Find all $x, y \in \mathbb{Z}_{\geq 0}$ s.t. $1 + 2^x = 3^y$. $(3,2)$, $(1,1)$

3 **Example.** Find all $x, y \in \mathbb{N}$ s.t. $3^x + 4^y = 5^z$.

1. When $y \geq 3$, $2^y \equiv 0 (8)$.
Then notice that $3^x \equiv 3, 1 \pmod 8$.
So $1 + 3^x \equiv 4, 2 \pmod 8$. No solution.
$\therefore$ $y = 0, 1, 2$.
When $y = 0$, $1 + 3^x = 1$. No solution.
When $y = 1$, $1 + 3^x = 2$, $x = 0$.
When $y = 2$, $1 + 3^x = 4$, $x = 1$.
$\therefore$ $(x, y) = (0,1), (1,2)$.

$1 + 2^x \equiv 1 (5)$.
$2^x \equiv 0 (5)$.
No solution.

2. When $x \geq 4$, $2^x \equiv 0 (16)$. Then $3^y \equiv 1 (16)$.
$\because$ $\text{ord}_{16}(3) = 4$, and so $4 | y$.
Let $y = 4a$, some $a \in \mathbb{Z}_{\geq 0}$. The equation becomes
$$1 + 2^x = 81^a.$$
Can consider mod 5.

$(9^a + 1)(9^a - 1) = 2^x$
$$\begin{cases} 9^a + 1 = 2^p & p, q \in \mathbb{Z}, \\ 9^a - 1 = 2^q, & p + q = x. \end{cases}$$

$2 = 2^p - 2^q$.

$p = 2$, $q = 1$. #

$\therefore$ $x = 0, 1, 2, 3$.
When $x = 0$, $3^y = 2$ ϕ
$x = 1$, $3^y = 3$, $y = 1$
$x = 2$, $3^y = 5$, ϕ
$x = 3$, $3^y = 9$, $y = 2$
$\therefore$ $(x, y) = (3,2), (1,1)$.

$$3^x + 4^y = 5^z, \quad x, y, z \in \mathbb{N}.$$

Mod 3, $\quad 0 + 1 \equiv (-1)^z \pmod 3$.

z even. Let $z = 2z_1$, some $z_1 \in \mathbb{N}$.

Then the equation becomes $\quad 3^x + 4^y = 5^{2z_1}$.

$$(5^{z_1} + 2^y)(5^{z_1} - 2^y) = 3^x.$$

However, $(5^{z_1} + 2^y, 5^{z_1} - 2^y) = (2(2^y), 5^{z_1} - 2^y)$

$$= 1$$

Checking;

$3^2 + 4^2 = 5^2 = 25$

$$\therefore \begin{cases} 5^{z_1} + 2^y = 3^x \\ 5^{z_1} - 2^y = 1 \end{cases}$$

Subtract, $2^{y+1} = 3^x - 1$.

$1 + 2^{y+1} = 3^x$.

From prev. example, rej.

$(y+1, x) = (3,2), (1,1)$

$(x, y) = (2,2)$.

Mod 3 again, $\quad (-1)^{z_1} + (-1)^y \equiv 0 \ (3)$

$(-1)^{z_1} - (-1)^y \equiv 1 \ (3)$.

$(-1)^{z_1} \equiv 2 (3), \quad (-1)^y \equiv 1 (3)$. $\quad z_1$ odd, y even.

From $5^{z_1} + 2^y = 3^x$, mod 4, $\quad 1 + 0 \equiv (-1)^x \pmod 4$.

x is also even. Let $x = 2X$, $X \in \mathbb{N}$. The equation becomes

$5^{z_1} + 2^y = 9^X$. Mod 8, if $y \geq 3$ $\quad 5^{z_1} + 0 \equiv 1 \pmod 8$.

z_1 even $\quad \#$.

$\therefore y = 2$. $\therefore 5^{z_1} - 2^2 = 1$, $z_1 = 1$, $z = 2$.

$x = 2$

$$\therefore (x, y, z) = (2, 2, 2).$$

Exercises

1. (AIME 2001) How many positive integer multiples of 1001 can be expressed in the form $10^j - 10^i$, where i and j are integers and $0 \leq i < j \leq 99$?

2. (IMO 1978) Let m and n be natural numbers such that $1 \leq m < n$. In their decimal representations, the last three digits of 1978^m are equal, respectively, to the last three digits of 1978^n. Find m and n such that $m + n$ has its least value.

3. (IMO 1990) Find all positive integers $n > 1$ such that $\dfrac{2^n + 1}{n^2}$ is an integer.

4. (Bulgaria 1996) Find all primes p, q s.t. $pq \mid (5^p - 2^p)(5^q - 2^q)$.

5. (USA TST 2003) Find all ordered prime triples (p, q, r) s.t. $p \mid q^r + 1$, $q \mid r^p + 1$, $r \mid p^q + 1$.

6. (China 2009) Find all primes p, q s.t. $pq \mid 5^p + 5^q$.

7. (China 2005) Find all $x, y, z, w \in \mathbb{Z}_{\geq 0}$ s.t. $2^x 3^y - 5^z 7^w = 1$.

8. (Bulgaria 2006) Find all $t, x, y, z \in \mathbb{N}$ s.t. $2^t = 3^x 5^y + 7^z$.

Extra Space